

Math 161

Midterm Exam 1 solutions

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1.

P	T	T	F	F
Q	T	F	T	F
not P	F	F	T	T
not Q	F	T	F	T
(not P) and Q	F	F	T	F
P and not Q	F	T	F	F
((not P) and Q) or (P and not Q)	F	T	T	F

Since the values of this truth table agree with that of “ P xor Q ,” the statements are logically equivalent.

2. We prove the identity by induction. For the base case $n = 1$, we simply observe that

$$1^2 = 1 = \frac{4 \cdot 1^3 - 1}{3}.$$

For the inductive step, assume that

$$1^2 + 3^2 + \cdots + (2n - 1)^2 = \frac{4n^3 - n}{3}$$

for some $n \in \mathbb{N}$. Then we have

$$\begin{aligned}
 1^2 + 3^2 + \cdots + (2n - 1)^2 + (2(n + 1) - 1)^2 &= \frac{4n^3 - n}{3} + (2n + 1)^2 \\
 &= \frac{4n^3 - n + 12n^2 + 12n + 3}{3} \\
 &= \frac{4n^3 + 12n^2 + 11n + 3}{3} \\
 &= \frac{4(n^3 + 3n^2 + 3n + 1) - n - 1}{3} \\
 &= \frac{4(n + 1)^3 - (n + 1)}{3}.
 \end{aligned}$$

Thus we have shown that the identity holds with n replaced by $n + 1$, as needed.

3. Since $B \neq \emptyset$, there exists $y \in B$, and by assumption $x \leq y$ for all $x \in A$, i.e. y is an upper bound for A . Thus A is bounded above, and since $A \neq \emptyset$, the least upper bound property says that A admits a least upper bound $\sup A$.

Similarly, the hypothesis $A \neq \emptyset$ implies that there exists $x \in A$, and $x \leq y$ for all $y \in B$, so x is a lower bound for B . Since $B \neq \emptyset$, the greatest lower bound property (proved on Problem Set 3) says that B admits a greatest lower bound $\inf B$.

For any $x \in A$ we have $x \leq y$ for all $y \in B$, i.e. x is a lower bound for B . Since $\inf B$ is the *greatest* lower bound for B , it follows that $x \leq \inf B$ for any $x \in A$. But this means that $\inf B$ is an upper bound for A , and since $\sup A$ is the *least* upper bound for A , we obtain $\sup A \leq \inf B$ as desired.